Comparison of various Time Complexities

$O(c) < O(\log \log n) < O(\log n) < O(n^{1/2}) < O(n) < O(n \log n) < O(n^2) < O(n^3) < O(n^k) < O(2^n) < O(n^n) < O(2^{2^n})$

$O(1)$

$O(2^n) \downarrow O(2^{n^2})$

$\log_2 1000$

1000×10

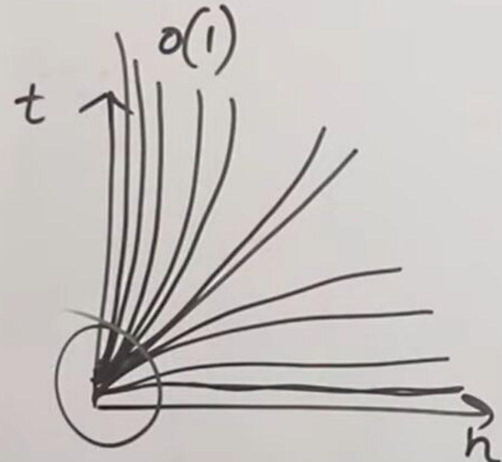
$4 < 10 < 33 < 1000 < 10000$

$K \geq 4$

$O(10^3)$ $O(10^9)$

1000)

$O(1)$ $O(1)$



$(10)^3$

1000

2^{10}

1000

Which term is dominating here

- Binary Search - $\log_2 n$ avg, worst
  - Best: 1, 2, 3, 4, 5 — $O(1)$
- Sequential Search - $O(n)$; $O(1)$; Avg $= n/2$
- *Quick Sort — $O(n^2)$ — $O(n \log n)$ Best, Avg.
  - worst $= O(n^2)$
  - 10, 9, 14, 6, 28
  - 1, 2, 3, 4, 5 $(n-1)$
- Merge Sort $\rightarrow O(n \log n)$
- *Insertion Sort $\rightarrow O(n^2)$ Avg, worst ; $O(n)$; $O(n)$
- Bubble Sort $\rightarrow O(n^2)$
- Heap Sort $\rightarrow O(n \log n)$
- Selection Sort $\rightarrow O(n^2)$

- Height of CBT $= O(\log n)$
- Insertion in Heap $(\log n)$; $O(1)$
- Construct Heap $(n \log n)$
- Delete from Heap $(\log_2 n)$
- Huffman $(n \log n)$
  - V = Vertex
  - E = Edges
- Prims Matrix $\rightarrow O(n^2)$, $O[(V+E)\log V]$ Heap
- Kruskal $O(E \log E)$
- DFS, BFS $O(V+E)$
- All pair Shortest $O(n^3)$; $O(n^3)$
- Dijkstra $O(V^2)$

That you have to remember the important points

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Qus. $f_1(n) = n^2 \log_{10} n$ ✓ $\quad f_2(n) = n(\log_{10} n)^{10}$

a) $f_1(n) = O\, f_2(n)$

b) $f_1(n) = \Omega\, f_2(n)$

$n = 16$

$(16)^2 \times 4 \qquad 16 \times (\log 16)$

$16 \times (4)^{10}$

$(10^9)^2 \log_{10} 10^9$

$\cancel{10^9} \times 10^9 \times 9$

$10^9 \times \cancel{9}$

$10^9 (\log_{10} 10^9)^{10}$

$\cancel{10^9} \times (9)^{10}$

$\cancel{9} \times 9^9$

$n = 10^9$

$n \cdot n\cancel{\log n} \qquad n\cancel{\log n}(\log n)^9$

$n \qquad (\log n)^9$

$\log n \qquad \log(\log n)^9$

$\log n \qquad \cancel{9} \cdot \log\log(n)$

$f_2(n) = f_1(n)$

$f_2(n) \leq c \cdot f_1(n)$

Means whatever option you get,
whether it is big O or omega,

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$(10^9)^2 \log_{10} 10^9$

$10^9 \times 9$

$10^9 \times \cancel{9}$

$10^9 (\log_{10} 10^9)^{10}$

$\cancel{10^9} \times (9)^{10}$

$\cancel{9} \times 9^9$

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$\log n \quad \log(\log n)^9$

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$f_2(n) \ldots$

you can check the terms here.

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$n \cdot n \cancel{\log n}$ $n \cancel{\log n} (\log n)^9$

n $(\log n)^9$

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Qus. $f_1(n) = 2^n$, $f_2(n) = n^{3/2}$, $f_3(n) = n\log_2 n$, $f_4(n) = n^{\log_2 n}$

Arrange in increasing order

$n = 16$ $\quad n = 256$

a) $f_2\ f_3\ f_4\ f_1$ $\quad 2^{16}\quad (16)^{3/2}\quad 16\times 4\quad (16)^4$

b) $f_2\ f_1\ f_3\ f_4$ $\quad n\log n < n^{3/2} < n^{\log n} < 2^n$ $\quad (256)^{3/2}\quad (256)\log_2(256)\quad (256)^{\log_2 256}$

c) $f_1\ f_2\ f_3\ f_4$ $\quad 2^{8\times 3/2}$

d) $f_3\ f_2\ f_4\ f_1$ $\quad \cancel{n}\, n^{1/2}\quad \cancel{n}\log n$ $\quad 2^{12}\quad 2^8\times 8\quad (256)^8$

$\sqrt{n}\quad \log n$ $\quad 2^8\times 2^3$

2^{11}

and find out the answer. Thank you.